There are six problems in total. You must solve the first five, and problem 6 is optional.

### Problem 1 (CO5): Pumping Lemma (5 points)

Consider the following language.

$$L = \{0^n 1^m : n = 3q + r_1, m = 3q + r_2, q \geq 0 \text{ and } 0 \leq r_1, r_2 \leq 2\}$$

**Prove** that L is not regular using the pumping lemma.

### Problem 2 (CO3): Context-Free Grammars (15 points)

Consider the following languages.

$$L_1 = \{w \in \{\texttt{a}, \texttt{b}\}^* : w \text{ is a palindrome}\}$$
$$L_2 = \{w \in \{\texttt{a}, \texttt{b}\}^* : \text{every second letter in } w \text{ is } \texttt{b}\}$$
$$L_3 = \{w \in \{\texttt{a}, \texttt{b}\}^* : w \text{ contains more } \texttt{a} \text{ than } \texttt{b}\}$$
$$L_4 = L_1 \cap L_2$$
$$L_5 = L_2 \cap L_3$$

Now solve the following problems.

(a) **Give** a context-free grammar that generates $\overline{L}_1$. (3 points) [Recall: $\overline{L}_1$ denotes the complement of the language $L_1$ i.e., $\overline{L}_1 = \Sigma^* - L_1$]

(b) **Give** a context-free grammar that generates $L_2$. (3 points)

(c) **Give** a context-free grammar that generates $L_3$. (3 points)

(d) **Write** all five-letter strings in $L_4$. (1 point)

(e) **Give** a context-free grammar that generates $L_4$. (2 points)

(f) **Give** a context-free grammar that generates $L_5$. (2 points)

(g) **Find** the total number of strings that belong to $L_5 \setminus L_1$. (1 point) [Recall: $L_5 \setminus L_1$ contains all strings that are in $L_5$ but not in $L_1$.]

### Problem 3 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Consider the following context-free grammar.

$$S \to P\texttt{0}Q$$
$$P \to \texttt{0}P\texttt{0} \mid \texttt{0}P\texttt{1} \mid \texttt{1}P\texttt{0} \mid \texttt{1}P\texttt{1} \mid \texttt{0} \mid \texttt{1}$$
$$Q \to \texttt{0}Q\texttt{1} \mid \texttt{1}Q\texttt{0} \mid \varepsilon$$

(a) **Give** a leftmost derivation for the string '1100101010'. (3 points)

(b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)

(c) **Show** that the given grammar is ambiguous by drawing two more parse trees (apart from the one that you have drawn in (b)) for the given string in (a). (4 points)

(d) **Write** two strings of length four that have exactly one parse tree in the given grammar. (1 point)





### Problem 4 (CO3): Pushdown Automata (10 points)

Consider the following languages.

$$L_1 = \{w \in \{0,1\}^* : w \text{ starts with 0 and ends with 1}\}$$
$$L_2 = \{w = 0^i 1^j 2^k : i, j, k \geq 0 \text{ and } k = 2i + j\}$$
$$L_3 = \{w_1\#w_2 : w_1, w_2 \in \{0,1\}^* \text{ and the number of 0s in } w_1 \text{ is equal to the number of 1s in } w_2\}$$

Now solve the following problems.

(a) **Give** the state diagram of a PDA that recognizes $L_1$. (3 points)

(b) **Give** the state diagram of a PDA that recognizes $L_2$. (3 points)

(c) **Give** the state diagram of a PDA that recognizes $L_3$. (4 points)

### Problem 5 (CO4): Turing Machines (10 points)

Consider the following languages.

$$L_1 = \{w_1\#w_2 : w_1, w_2 \in \{0,1\}^* \text{ and } \text{count}(w_1, 0) = \text{count}(w_2, 0)\}$$
$$L_2 = \{w\#w' : w \in \{0,1\}^*\}$$

(a) For the configuration $011q_4010$ of a Turing Machine, **what** is the current position of the tape head? (1 point)

(b) **Give** the state diagram of a turing machine that decides $L_1$. [Note: $\text{count}(w, a)$ denotes the number of occurrences of the symbol $a$ in the string $w$.] (3 points)

(c) **Give** the state diagram of a turing machine that decides $L_2$. [Note: $w'$ denotes the bitwise complement of $w$ (obtained by flipping each bit: $0 \mapsto 1$, $1 \mapsto 0$), for example, if $w = 01101$, then $w' = 10010$.] (4 points)

(d) **Prove** that the collection of Turing decidable languages is closed under intersection. (2 points)

### Problem 6 (Bonus): Turing Machines (5 points)

Let $HALT_{TM} = \{\langle M, w\rangle : M \text{ is a TM and } M \text{ halts on input } w\}$. Given that $HALT_{TM}$ is undecidable, **prove** that the collection of Turing recognizable languages is not closed under complement.

After you are finished with the exam, please indicate where you stand on the smiley face spectrum.